Logan Schexnaydre

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Education

Michigan Technological University Houghton, MI
PhD, Computer Engineering (Expected Graduation: Summer 2026) *Jan 2022 — Present*

- Cumulative GPA: 4.0/4.0
- Relevant Coursework: Machine Learning, Graduate Introduction to Robotics, Probability and Stochastic Processes, Detection and Estimation Theory, Linear Systems

Michigan Technological University Houghton, MI
Bachelor of Science, Computer and Electrical Engineering *Aug 2017 — Dec 2021*

- Cumulative GPA: 3.86/4.0

Research and Educational Experience

Graduate Assistant May 2022 — May 2024, May 2025 — Present
AutoDrive II Competition, Michigan Technological University *Houghton, MI*

- Contributed to 10 team awards won over four years through hands-on mentorship of undergraduate students in several autonomous vehicle design challenges.
- Mentored a simulation-based team (4-7 undergraduate students) tasked with testing and verifying the functionality of perception, path planning, and control systems in simulation using MathWorks tools.
- Integrated autonomous vehicle subsystem sensors with a localization stack using C++, ROS, coordinate transforms, and path planning tools used in a custom autonomous vehicle during physical competition events.

Graduate Research Assistant May 2022 — August 2025
NEXTCAR II Research Program, Michigan Technological University *Houghton, MI*

- Implemented C++ lidar processing code for reducing energy consumption by 20% in autonomous vehicle following.
- Developed a lidar-based road surface profiling system to provide a lookahead signal to a control system for increasing ride safety and comfort.
- Evaluated the performance of lidar-based processing systems through Python and stochastic models in MATLAB.

Undergraduate Researcher Jun 2018 — Aug 2018
Oakland University, Department of Electrical and Computer Engineering *Auburn Hills, MI*

- Designed and completed an experiment to compare the ability of IMU and flex sensors to measure knee flexion.
- Collected and processed sensor data in real-time through Arduino and MATLAB.

Engineering Experience

Component and Hardware Systems Engineering Intern August 2020 — August 2021
National Aeronautics and Space Administration, Goddard Space Flight Center *Greenbelt, MD*

- Implemented C++ unit tests for a small satellite GNSS component.
- Organized the wire harness for the EGSE of the Roman Space Telescope (RST) deployment, propulsion, and GSE subsystems.
- Automated safe-to-mate testing of the RST deployable, propulsion, and GSE wire harness connectors.

Software Engineering Intern May 2020 — August 2020
Space Dynamics Laboratory *Logan, UT*

- Developed parameter selection hooks and interface for testing scoring algorithms for optimizing the task allocations of small satellites within a MATLAB-based satellite simulation software.

Guidance, Navigation, and Controls Intern May 2019 — August 2019
National Aeronautics and Space Administration, Wallops Flight Facility *Chincoteague, VA*

- Analyzed rocket/spacecraft trajectories for system testing scenarios through Python scripting on Linux and Windows.
- Wrote and completed subsystem and unit-level tests for avionics components.
- Performed component testing and wire-harness troubleshooting for balloon science missions.

Payload Team Lead - Auris Mission January 2018 — December 2021
Michigan Tech Aerospace Enterprise *Houghton, MI*

- Led an independent team of up to 10 students to develop hardware and software components for a spacecraft payload.
- Developed and documented embedded software (Python, C/C++) from scratch that executes payload operations.
- Tested Software Defined Radios and GNSS boards using Linux and scripting software.

Publications

L. Schexnaydre, I. Q. Mattson, C. Cornwall, and J. P. Bos, "Modeling the Effects of Residual Snow Clutter on Position Estimation and Vehicle Energy Efficiency," **Presented at 2026 IEEE International Workshop on Metrology for Automotive (MetroAutomotive) by J. P. Bos**, Brescia, Italy, Jun. 2026.

C. Cornwall, C. Majhor, **L. Schexnaydre**, I. Mattson, and J. P. Bos, "Can We Get There Faster: Tuning Sample-Based Path Planners," IEEE Robotics and Automation Letters, vol. 11, no. 6, pp. 7238–7245, Jun. 2026, doi: 10.1109/LRA.2026.3685923.

L. Schexnaydre, A. Poovalappil, D. Robinette, and J. Bos, "A Comparison of Road Grade Preview Signals from Lidar and Maps," SAE International, Apr. 2026. doi: 10.4271/2026-01-0026. **Presented at 2026 WCX SAE World Congress Experience.**

I. Q. Mattson, **L. Schexnaydre**, and J. P. Bos, "Red versus Infrared: comparing 900nm and 1550nm Lidar performance in arctic winter conditions," in Laser Communication and Propagation through the Atmosphere and Oceans XIV, SPIE, Sep. 2025, p. 1361702. doi: 10.1117/12.3064732.

A. Poovalappil A. Robare, **L. Schexnaydre**, P. Santhosh, M. Bahramgiri, J. P. Bos, B. Chen, J. Naber, D. Robinette, "On-Road Investigation of Energy Saving Opportunity for Autonomous Light-Duty Vehicles through Automated Vehicle-Following in Safe Distance Scenarios," presented at the WCX SAE World Congress Experience, SAE International, Apr. 2025. doi: 10.4271/2025-01-8029.

M. H. Schmelzle, **L. Schexnaydre**, N. Spike, D. Robinette, and J. Bos, "Facilitating Project-Based Learning Through Application of Established Pedagogical Methods in the SAE AutoDrive Challenge Student Design Competition," presented at the WCX SAE World Congress Experience, SAE International, Apr. 2024. doi: 10.4271/2024-01-2075.

L. Schexnaydre, A. Poovalappil, M. Schmelzle, D. Robinette, and J. P. Bos, "Using automated vehicle positioning to improve efficiency in vehicle platooning," in Autonomous Systems: Sensors, Processing, and Security for Ground, Air, Sea, and Space Vehicles and Infrastructure 2023, SPIE, Jun. 2023, pp. 202–211. doi: 10.1117/12.2664430.

Presentations and Posters

Guest Lecture, EE5531: Graduate Introduction to Robotics, Michigan Tech, Mar. 23, 2026

- Topic: Adaptive Monte Carlo Localization

L. Schexnaydre, "Efficient Perception Algorithms Can Save Energy for Autonomous Vehicles," poster presented at the Graduate Research Colloquium, Michigan Technological University, Mar. 26, 2024.

L. Schexnaydre and A. Burr, "Tracking Lower Limb Movement Using an Integrated Sensor Approach," poster presented at the Mid-Michigan Symposium for Undergraduate Research Experiences, Michigan State University, Jul. 24, 2018.

Outreach and Service

Vice Chair January 2026 — April 2026
Michigan Tech IEEE Robotics and Automation Society Student Chapter *Houghton, MI*

Skills

- **Programming Languages**: C/C++, Python, MATLAB
- **Technologies**: Git, Linux, Bitbucket, GitHub, LaTeX, ROS, Point Cloud Library, Microsoft Office